

India's Space Software Ecosystem

Space Atlas — Final Project Report

A structured analysis of the software systems, workflows, users, quality considerations, communication approach, localization, accessibility, and final Space Atlas direction developed during the internship.

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1. Executive Summary

This report presents the work completed during the Agnirva Software Internship on India's Space Software Ecosystem. The project was treated as a knowledge and ecosystem analysis rather than a simple list of software tools. The main purpose was to organize complex information about the software layer supporting space activities and make it easier for students, beginners, educators, and other interested learners to understand.

The work progressed through six connected artifacts. The BRD defined the user need and project boundaries. The RLD mapped the software landscape and identified gaps and evidence-sensitive claims. The VQRD established quality and validation principles. The PRD converted findings into a practical knowledge-product direction. The SCRP addressed stakeholder communication. The LAAP addressed localization and accessibility while preserving technical meaning.

The final direction is a Space Atlas knowledge product connecting software areas, workflows, users, evidence, communication, localization, and accessibility. A public SpaceAtlas project repository was also created to represent this work digitally.

Key outcome: a structured, learner-oriented Space Software Ecosystem analysis with a defined approach to reliability, communication, localization, accessibility, and final digital presentation.

2. Project Background and Purpose

Space activities depend on interconnected software systems. Software can support spacecraft operations, ground-side processing, navigation and positioning, data handling, simulation, monitoring, analytics, and user-facing services. Understanding this layer requires more than identifying individual tools because the systems interact through workflows.

Information about space technology may be distributed across official websites, technical documents, research publications, and other sources. Terminology and system relationships can also be difficult for new learners to follow. The project therefore focused on creating a structured knowledge approach that explains the ecosystem in simple language while maintaining source awareness and technical accuracy.

Project purpose

- Organize major software areas connected with space activities.
- Explain relationships between spacecraft, ground systems, data, operations, analysis, and users.
- Provide a clear learning path for students and beginners.
- Use reliable sources and distinguish facts from interpretation.
- Include quality, communication, localization, and accessibility considerations.

3. Objectives and Scope

Objectives

- Identify major areas of the Space Software Ecosystem.
- Understand how software participates in space-related workflows.
- Identify information needs of students, beginners, educators, and other stakeholders.
- Create a practical structure for a readable knowledge product.
- Define evidence and quality checks for important technical claims.
- Develop communication, localization, and accessibility guidance.

Scope and boundaries

- The project focuses on ecosystem understanding and knowledge organization rather than building mission-critical flight software.
- Advanced AI systems, complex integrations, and highly specialized tools were kept outside the initial scope unless needed for explanation.
- The project does not claim access to confidential mission systems or unpublished technical information.
- Where public information is incomplete, the gap is treated as a limitation rather than evidence of absence.

4. Space Software Ecosystem Overview

The ecosystem can be understood as several connected software layers. Each layer has a different role, but the value comes from how these layers work together.

Area	Role in the ecosystem
Flight software	Supports onboard control, telemetry, command handling, and interaction with spacecraft subsystems.
Ground systems	Supports communication, telemetry acquisition, command operations, monitoring, and mission-related data handling.
Navigation & positioning	Connects satellite signals, processing, positioning, and timing-related services.
Data processing	Transforms received or collected data into organized information suitable for analysis and applications.
Simulation & testing	Supports development, validation, training, and preparation through controlled software environments.
Monitoring & dashboards	Presents system status, telemetry, alerts, and operational information to users.
Analytics	Helps users examine processed information and identify patterns or useful insights.
Applications	Turns processed information and services into outputs that can be used by specific users.

Conceptual ecosystem workflow

Stage	Flow	Stage	Flow	Stage
Space assets	→	Communication	→	Ground systems
Ground systems	→	Data processing	→	Analytics
Analytics	→	Applications	→	Users

5. Users and Stakeholders

The project considered different audiences because the same technical information may need to be explained differently depending on the user's background and purpose.

Audience	Need / consideration
Students & beginners	Simple explanations, definitions, clear workflows, examples, and an easy learning path.

Audience	Need / consideration
Regional-language learners	Language support while preserving technical terms and original meaning.
Educators & professors	Reliable sources, structured material, and clear distinction between facts and interpretation.
Researchers / technical learners	References, evidence, deeper resources, and transparent limitations.
Project partners / editors	Clear communication, review processes, and consistent content standards.
Mobile / accessibility users	Readable text, responsive layout, clear navigation, contrast, and accessible interaction.

6. Research and Landscape Analysis (RLD)

The RLD stage examined the wider Space Software Ecosystem using public and trusted reference categories, including ISRO, IN-SPACe, Agnirva project material, government sources, and technical or research documentation.

Research landscape

- Mission planning and operations.
- Flight software and onboard systems.
- Ground systems and communication support.
- Satellite data processing and information workflows.
- Simulation and testing.
- Monitoring, dashboards, and analytics.
- Navigation and positioning-related software and data.
- User-facing applications and services.

Key findings

- Space software appears across many activities rather than existing as a single software category.
- Ground systems are important for acquisition, processing, monitoring, communication, and operational support.
- Navigation examples such as NavIC demonstrate relationships between space infrastructure, ground systems, software, and data.
- Public information does not always provide a complete end-to-end workflow or complete list of tools; missing public information should not automatically be treated as evidence of absence.

Research priorities

- Software categories and their purpose.
- Data and operational workflows.
- Tools and platforms appropriate for student learning where evidence is available.
- Reliable sources that can support important technical statements.

Evidence-sensitive claims

- Claims that a particular software product is the 'best'.
- Claims that a specific tool was used in a particular mission without a reliable source.
- Performance or accuracy claims without supporting evidence.
- Future technology or timeline claims without authoritative documentation.

7. Quality, Validation and Reliability (VQRD)

The VQRD stage established how the project should protect itself from incorrect, outdated, unclear, unsupported, or exaggerated information. The goal was to define how important claims should be reviewed before publication.

Quality risks

- Incorrect or outdated information.

- Unsupported technical claims.
- Exaggerated statements or rankings.
- Confusion between fact and interpretation.
- Unclear or ambiguous technical explanations.

Evidence approach

- Prefer official organizations, government sources, research papers, and technical documentation for important claims.
- Treat blogs, social media, and secondary summaries cautiously unless the underlying claim can be checked.
- Narrow a claim when the available evidence supports only a narrower statement.
- Remove unsupported or misleading claims instead of filling evidence gaps with assumptions.

Validation workflow

Step 1	Step 2	Step 3	Step 4	Step 5
Claim	Source	Evidence	Review	Publish

8. Product Requirements and Proposed Knowledge Structure (PRD)

The PRD converted the research and quality findings into a proposed knowledge-product structure. The primary use case is to help students and beginners find a topic, understand its purpose and workflow, and reach reliable resources.

Core features

- Organized ecosystem categories.
- Search or topic discovery.
- Clear topic explanations.
- Workflow-oriented learning.
- Resources and references.
- Simple, responsive presentation suitable for mobile users.

Proposed user flow

1	2	3	4	5
Open platform	Explore / Search	Topic explanation	Workflow	Sources / Resources

Constraints and acceptance criteria

- Reliable sourcing remains a core requirement.
- The experience should remain understandable to beginners without removing essential meaning.
- The first version should stay focused instead of attempting advanced AI or complex integrations.
- Users should be able to find a topic and understand its basic workflow.
- Content should be checked for claims and presented in a responsive layout.

9. Stakeholder Communication Plan (SCRP)

The SCRП stage focused on how the Space Software Ecosystem should be communicated to different stakeholders. The central principle was to adapt the message to the audience while keeping claims supported by evidence.

Element	Plan
Objective	Make the ecosystem understandable, useful, and credible.
Core message	Software supports flight operations, ground systems, navigation, data processing, simulation, monitoring, and analytics.
Primary channels	Knowledge product / web platform, technical documents, reports, presentations, and briefings.
Proof points	Official sources, research papers, technical documentation, Agnirva material, and project artifacts.
Main risks	Overclaiming, weak evidence, confusing language, and unclear audience targeting.

Element	Plan
Response approach	Answer with evidence, clarify uncertainty, explain value without exaggeration, and acknowledge limitations.

10. Localization and Accessibility Action Plan (LAAP)

The LAAP stage focused on making technical knowledge easier to access for people with different language backgrounds, technical familiarity, and device or accessibility needs. Localization was treated as more than direct translation: it includes understandable explanations, familiar examples, readable layouts, and preservation of technical meaning.

Audience adaptations

Audience	Barrier	Adaptation
Students / beginners	Limited technical background and complex terminology.	Simple English, definitions, clear sections, examples, and basic workflows.
Regional-language learners	English-only technical information.	Kannada / Hindi support where appropriate, careful translation, and preservation of technical terms.
Mobile / accessibility users	Small text, poor contrast, crowded layouts, unclear navigation.	Responsive design, readable typography, contrast, labels, spacing, and accessible navigation.

Accessibility and localization rules

- Use learner-friendly language without removing important technical meaning.
- Compare original and translated content to prevent mistranslation.
- Preserve technical terms and workflows when translation could change their meaning.
- Use headings, short paragraphs, readable fonts, sufficient contrast, labels, spacing, and responsive layouts.
- Use human review for technical accuracy, language quality, regional suitability, examples, and accessibility.

Practical access tests

- Beginner understanding test: the learner should be able to explain the purpose and key terms.
- Translation test: the localized version should preserve the technical meaning of the original.
- Device test: users should be able to read, navigate, and access information on mobile and desktop layouts.

11. Project Journey: BRD to Space Atlas

The six artifacts were treated as connected stages of one project. Each artifact answered a different question and prepared the foundation for the next stage.

Week	Artifact	Main contribution
2	BRD	Defined the user need, requirements, boundaries, trust principles, and success measures.
3	RLD	Mapped the ecosystem, research findings, information gaps, priorities, and risky claims.
4	VQRD	Defined quality risks, evidence standards, rejection rules, and review checks.
5	PRD	Defined the use case, features, user flow, constraints, acceptance criteria, and launch readiness.
6	SCRP	Defined stakeholder groups, messages, channels, proof points, risks, and response approach.
7	LAAP	Defined audience barriers, localization choices, accessibility rules, tests, and human review.

Week	Artifact	Main contribution
8	Space Atlas	Brings research, product thinking, communication, quality, localization, and accessibility into one visible knowledge project.

12. Space Atlas Website

To make the internship work visible as practical work, a public project repository named **SpaceAtlas** was created. The website is intended to turn the knowledge structure developed during the internship into an accessible digital experience.

Project repository

<https://github.com/suhas-s-pai/SpaceAtlas>

Website direction

- Dark, premium, space-inspired visual design.
- Cinematic visual treatment with restrained space-themed animation.
- Sections for ecosystem areas, workflows, research, quality, communication, accessibility, and sources.
- Responsive design for desktop, tablet, and mobile users.
- A project journey showing the connection between BRD, RLD, VQRD, PRD, SCRP, and LAAP.
- Clear distinction between documented information, conceptual examples, and future possibilities.

Content responsibility

The website should not present conceptual diagrams, example dashboards, or future ideas as live mission systems or verified operational data. Claims should remain linked to appropriate evidence, and future features should be labeled as possibilities rather than completed capabilities.

Expected user journey

- Enter the Space Atlas and understand its purpose.
- Explore major software ecosystem areas.
- Follow a conceptual workflow from space assets to user-facing services.
- Read research and quality guidance.
- Understand localization and accessibility choices.
- Use source references to continue learning.

13. Security, Reliability and Responsible Communication

Because space software is associated with technically important systems, the knowledge product should avoid presenting uncertain information as fact. The project therefore treats reliability and responsible communication as content requirements.

Key considerations

- Use trusted and relevant sources for important technical claims.
- Do not publish sensitive, confidential, or restricted operational information.
- Do not expose credentials, private data, internal system details, or security-sensitive information.
- Clearly label conceptual diagrams and examples.
- Record limitations where public information is incomplete.
- Review content when important sources or technical information changes.

- Keep facts separate from interpretation and opinion.

14. Final Synthesis

The project moved from understanding a user need to designing a structured knowledge solution. The BRD established why the project was needed. The RLD explained the landscape. The VQRD created a reliability framework. The PRD shaped the knowledge product. The SCRП made the information communicable to different stakeholders. The LAAP extended the approach to language, accessibility, and wider usability.

Together, these stages create a practical model for presenting India's Space Software Ecosystem to learners. The central idea is that software should be understood as an interconnected layer connecting space assets, communications, ground systems, data, analysis, operations, and applications.

The resulting Space Atlas direction is therefore not simply a catalogue. It is a structured learning and knowledge framework that combines ecosystem mapping, workflow explanation, evidence awareness, product structure, communication planning, and accessibility.

Project outcome: a documented and structured Space Software Ecosystem analysis, supported by six project artifacts and represented through the SpaceAtlas digital project.

15. Future Possibilities

The current project provides a foundation that could be extended in future iterations. These are possibilities rather than completed features.

- Interactive ecosystem maps connecting software areas and workflows.
- Expanded educational resources for different learning levels.
- Additional regional-language support with human review.
- Interactive workflow visualizations.
- A searchable knowledge base with stronger source management.
- Structured learning modules and guided topic paths.
- Additional visualization tools for explaining complex relationships.
- Periodic content review to maintain source quality and relevance.

16. Conclusion

The Agnirva Software Internship provided a structured opportunity to study India's Space Software Ecosystem from multiple professional perspectives. The work covered requirements analysis, research, validation, product planning, stakeholder communication, localization, and accessibility.

The final Space Atlas concept brings these perspectives together into one coherent knowledge project. Its objective is to help learners understand how software contributes to space capability without oversimplifying technical relationships or making unsupported claims.

The project also demonstrates a practical software-product workflow: define the problem, research the landscape, validate information, shape a product, plan communication, improve accessibility, and synthesize the work into a usable output.

17. Project Evidence and References

The report is based primarily on the project artifacts prepared during the Agnirva internship and on the public-source categories identified during the research stage.

Project artifacts

- Business Requirements Document (BRD) — Week 2.
- Research and Landscape Document (RLD) — Week 3.
- Validation and Quality Requirements Document (VQRD) — Week 4.
- Product Requirements Document (PRD) — Week 5.
- Stakeholder Communication and Response Plan (SCRP) — Week 6.
- Localization and Accessibility Action Plan (LAAP) — Week 7.
- SpaceAtlas project repository — <https://github.com/suhas-s-pai/SpaceAtlas>

Public source categories used for research

- Indian Space Research Organisation (ISRO) public information.
- IN-SPACe public information.
- Government and official technical sources.
- Research papers and technical documentation.
- Agnirva internship project material and instruction resources.

Report note

This report describes the work completed as part of the Agnirva Software Internship project. Where a technical detail was not publicly established in the project research, it has been presented at a general ecosystem level rather than as a specific operational claim.

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